



Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's Program/PhD
Course
End-Semester Examination-Autumn 2025

Subject: Molecules of life
Full Marks: 50

Subject Code(s): BIS1101
Time Allotted: 3 h

Answer Part A and Part B separately. Where a question has multiple sub-parts, answer them together.
Present all answers in the order of the question numbers, starting from the lowest.

Part A

Q1: MCQ (choose one correct answer, 1/2 marks for each correct answer)

i) Which step occurs last in the functioning of a CCD camera?

- A. Photons strike the sensor and generate electron-hole pairs
- B. Charge packets are shifted through potential wells
- C. Charge is read out and converted to voltage
- D. Electrons are accumulated in each pixel during exposure

ii) What is the primary role of the CCD (Charge-Coupled Device) in a camera?

- A. To amplify the light signal directly
- B. To convert incoming light into electrical charge
- C. To store digital images
- D. To provide color information using filters

iii) Which optical component in a DIC microscope splits the incoming light into two orthogonally polarized beams?

- A. Condenser lens
- B. Wollaston or Nomarski prism
- C. Polarizing filter in the eyepiece
- D. Beam splitter cube

iv) In DIC microscopy, the two split beams pass through the specimen with a slight spatial offset. What does this offset achieve?

- A. It increases the numerical aperture of the system

- B. It allows the beams to acquire phase differences based on refractive index gradients
- C. It prevents photobleaching of the sample
- D. It ensures color separation

v) How is contrast generated in DIC images?

- A. By converting differences in absorption into intensity variations
- B. By converting differences in fluorescence emission into color
- C. By interfering the two beams after they recombine, converting phase shifts into intensity differences
- D. By blocking out-of-focus light with a pinhole

vi) In phase contrast microscopy, what does the phase plate located in the objective lens do?

- A. Amplifies the light intensity
- B. Converts phase differences into color contrast
- C. Shifts the phase of the direct (undeviated) light
- D. Blocks scattered light from the sample

vii) Which statement best differentiates Phase Contrast from DIC microscopy?

- A. Phase Contrast uses interference of polarized light, while DIC uses absorption of light.
- B. Phase Contrast shifts the phase of direct light using a phase plate, while DIC uses

birefringent prisms to split and later recombine polarized light.

C. DIC relies on fluorescence emission, while Phase Contrast relies on scattering.

D. Both techniques use the same optical components but produce different colors

viii) Which parameter most directly improves optical resolution in a light microscope?

A. Increasing camera exposure time

B. Increasing numerical aperture (NA)

C. Increasing working distance

D. Using a thicker coverslip

ix) A microscope provides a resolution of 300 nm at 600 nm illumination.

What will be the resolution at 450 nm, assuming NA is unchanged?

A. 225 nm

B. 300 nm

C. 450 nm

D. 150 nm

x) Calculate the resolution for Air (NA = 0.95) and Oil (NA = 1.40) objectives at 600 nm wavelength. Which pair is correct?

A. Air: 385 nm; Oil: 261 nm

B. Air: 600 nm; Oil: 385 nm

C. Air: 261 nm; Oil: 385 nm

D. Air: 700 nm; Oil: 500 nm

Q2 What is meant by a lipid bilayer phase transition? (1.5 marks)

Q3 How does cholesterol influence the phase behavior of lipid membranes? (1.5 marks)

Q4 What factors determine the phase transition temperature (T_m) of a lipid bilayer? (1.5 marks)

Q5 What happens to membrane permeability and fluidity during the gel-to-liquid phase transition? (1.5 marks)

Q6 Describe the molecular characteristics of liquid-ordered (L_o) and liquid-disordered (L_d) phases in lipid membranes. How do these phases influence membrane protein distribution and function? (4 marks)

Q7 Compare the processes of DNA replication and transcription in terms of their purpose, enzymes involved, template usage, directionality, initiation mechanisms, elongation steps, fidelity, and termination. Discuss how the products of these two processes differ and explain why their regulation is essential for proper cellular function. (5 marks)

Part B

Answer any six questions:

1. Name the cells which have large and numerous Golgi apparatus. What is the difference between O-linked glycosylation and N-linked glycosylation? What is the role of COPI vesicles? (1+3+1)
2. What is the glycocalyx of the lysosomes? What are the key functions of lysosomes? What is the Autolysosome? (1+3+1)
3. Mention the major functions of smooth endoplasmic reticulum (SER). How SER involved in glycogenolysis? (3+2)
4. What is the translocon of the endoplasmic reticulum? How are soluble proteins targeted to the ER lumen? How to induce ER stress in cells? (2+2+1)
5. What are the major functions of vacuole in animal cell? Why plant cells have larger vacuoles? Do animal cells have contractile vacuoles? What is the effect of accumulation of vacuoles in skeletal muscle cells? (2+1+1+1)
6. What causes mitochondria permeability transition pore (mPTP) to open? How ER controls Mitochondrial fission and fusion? When Mitochondrial nanotunnels forms? (1+3+1)
7. Mention the three major pathways of mitochondrial quality control. What are mitochondrial precursor proteins or preprotein? How do they export to mitochondria. (3+1+1).

8. Name some enzymes found in peroxysome? Cellular metabolism produces hydrogen peroxide, which is toxic, how this is primarily overcome by the mammalian cells? What is the importance of beta-oxidation in peroxisome? Why Peroxisomes are crucial for seed germination?

(1+2+1+1)

9. What is cotranslational and post-translational protein translocation? Outline one experiment to show the significance of signal sequences? What happens to a protein which does not have any signal sequence?

(2+2+1)

10. Mention the role of BiP (binding immunoglobulin protein), a major endoplasmic reticulum (ER) chaperone. What is Endoplasmic-reticulum-associated protein degradation (ERAD)? How ERAD is involved in disease-Give one example.

(2+2+1)